

FRACTAL-WAVE ALGEBRA (FWA): FORMAL PUBLICATION DRAFT

A Paradigm Shift from Static Projections to Emergent Wave Dynamics

Abstract

Fractal-Wave Algebra (FWA) introduces a dynamic mathematical ontology where numbers represent quantum states of sequential symmetry breaking, and algebraic operations represent directed transitions between evolving wave configurations. Classical arithmetic emerges as a degenerate projection of dynamic wave processes when local symmetry deformation approaches zero. FWA provides a unified framework for dynamic arithmetic, wave-native computation, and photonic AI architectures. The theory resolves classical paradoxes such as P vs NP and the Riemann Hypothesis through wave interference, fractal resonance, and symmetry-stability constraints. This publication formalizes the ontology, axioms, operators, equations, computational model, hardware blueprint, and experimental roadmap for FWA.

1. Introduction

Classical mathematics treats numbers as static abstractions and operations as timeless relations. This static ontology fails to describe dynamic systems, wave interactions, and non-linear computational processes. FWA replaces static arithmetic with dynamic wave evolution, where each algebraic expression is a projection of underlying symmetry-breaking transitions. The motivation is to unify mathematics, physics, and computation under a single wave-based algebraic framework.

FWA builds on prior art in wave computing, photonic integration, and fractal analysis, establishing a new paradigm for AI, hardware, and mathematical theory.

2. Ontological Foundations of FWA

2.1 Core Principles

- **Number** — quantum of state; record of sequential symmetry-breaking events.
- **Zero Node (Z)** — perfectly balanced baseline state.
- **Symmetry Breaking (AS)** — generator of dynamic shifts.
- **Operation** — directed transition between wave states.
- **State** — configuration of wave symmetry at a given moment.

- **Projection** — classical arithmetic as collapsed wave behavior.

2.2 Formal Definitions

Axiom 1. Number n is a finite sequence of symmetry-breaking quanta:

$$n = \sum_{i=1}^k AS_i$$

Axiom 2. Zero Node Z satisfies:

$$\forall AS : Z + AS = AS$$

Axiom 3. Algebraic operation is a directed wave transition:

$$a \rightarrow b = O(a, AS)$$

Axiom 4. Classical arithmetic emerges when:

$$AS \rightarrow 0$$

3. Dynamic Arithmetic

3.1 Static vs Dynamic

Classical arithmetic:

$$a + b = c$$

is a static projection of dynamic evolution:

$$Da > (0+) > Pc$$

where Da is the deformation amplitude, Pc is the projected classical result.

3.2 Collapse to Classical Arithmetic

When symmetry deformation vanishes:

$$\lim_{AS \rightarrow 0} O(a, AS) = a + b$$

Thus classical arithmetic is a macro-illusion of collapsed wave dynamics.

4. Wave-Based Computational Model

4.1 Multi-Layered Wave Interference

Computation occurs through fractal wave layers:

$$W = \{W_1, W_2, \dots, W_k\}$$

True solutions reinforce resonance; false paths dampen.

4.2 Complexity Reinterpretation

NP problems mapped to wave spectra collapse to polynomial wave complexity:

$$O(n^2)$$

because interference eliminates exponential branching.

5. FWA Interpretation of Mathematical Paradoxes

5.1 P vs NP

Von Neumann hardware computes sequentially → exponential NP complexity.

Wave hardware computes via interference → polynomial collapse.

Thus:

$$P = NP \text{ (in wave domain)}$$

5.2 Riemann Hypothesis

Non-trivial zeros of $\zeta(s)$ are fractal nodes on a wave boundary.

Deviation from $\text{Re}(s) = 1/2$ breaks symmetry and collapses analytic stability.

Thus all zeros lie on the critical line.

6. Hardware Blueprint for Wave Computing

6.1 Photonic Integration Circuits

Light replaces electrons. Information encoded in phase, amplitude, and frequency. Metamaterials enable instantaneous superposition.

6.2 In-Memory Wave Processing

Wave medium stores and processes data simultaneously. Quartz crystals, plasma cells, harmonic resonance.

6.3 Wave-AI Architecture

Matrix weights replaced by phase-resonance maps. Errors = geometric variations across 7 AS levels.

Self-healing via resonance stabilization.

7. Mathematical Framework

7.1 Operators

• Deformation operator:

$$O(a, AS) = a + \Delta_{AS}$$

• Transition operator:

$$T(a, b) = O(a, AS_b)$$

- Symmetry gradient:

$$\nabla_s = \partial \mathbf{AS} / \partial \mathbf{x}$$

7.2 Equations

Wave evolution:

$$W_{t+1} = W_t + O(W_t, AS_t)$$

Fractal recursion:

$$F_{n+1} = 1/k \sum_{i=1}^k F_i$$

Collapse condition:

$$\text{If } AS \rightarrow 0 \Rightarrow W \rightarrow \textit{classical arithmetic}$$

8. Computational Framework

8.1 Fractal-Wave Processing Algorithm

Recursive decomposition into wave layers. Local averaging stabilizes volatility. Fractal recombination yields final state.

8.2 Prior Art Code

```
def self_similar_acceleration(data_stream, layers=4):
    if len(data_stream) < layers:
        return sum(data_stream) / max(len(data_stream), 1)

    chunk = len(data_stream) // layers
    wave_layers = []

    for i in range(layers):
        sub_grid = data_stream[i * chunk : (i + 1) * chunk]
        wave_layers.append(sum(sub_grid) / max(len(sub_grid), 1))

    return sum(wave_layers) / len(wave_layers)
```

9. Applications

- Wave-AI
- Non-linear data processing
- Quantum-adjacent computing
- Cryptography
- Mathematical analysis
- Photonic hardware

- Plasma computing

10. Experimental Roadmap

- PIC prototypes
- Plasma wave cells
- NP wave benchmarks
- $\zeta(s)$ resonance mapping
- AS-level error diagnostics

11. Discussion

FWA unifies wave physics and algebra. Limitations include hardware availability and full formalization. Future work involves a deeper operator theory, hardware scaling, and comprehensive AI integration.

12. Conclusion

FWA establishes a dynamic algebraic paradigm where arithmetic emerges from wave evolution. It resolves classical paradoxes, defines wave-native computation, and provides a clear blueprint for future hardware architectures. This publication formalizes the theory and sets a solid foundation for further international development.

Appendices

- A — Full axiom set
- B — Extended equations
- C — Wave transition diagrams
- D — Hardware schematics
- E — Prior art registry

Footnote: This is for informational purposes only. For medical advice or diagnosis, consult a professional. AI responses may include mistakes.